

Dynamic Modelling for Human-centered Systems

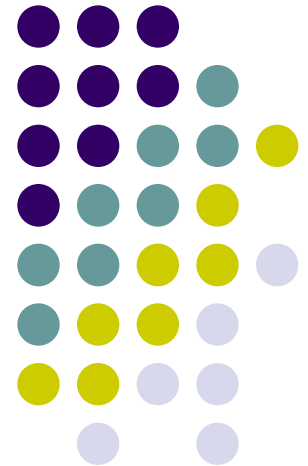
February – March 2025

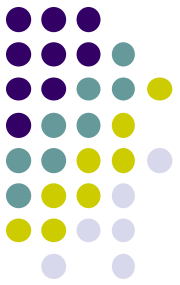
Michel Klein

Nimat Ullah

TA's:

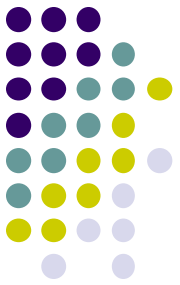
Sofronia, Nane, Ella, Rushil, Yash, Damla, Atilla,
Orkun, Diana, Kevin, Ana, Sofija





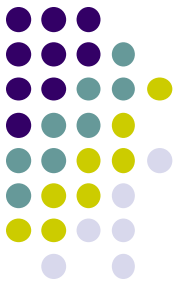
Why this course?

- Processes in the real world can be described and simulated by computers
- Simulations can be used to better understand and to make predictions
- Computer systems can use such models to behave intelligently
 - especially when knowledge about human functioning is used in for interaction with humans



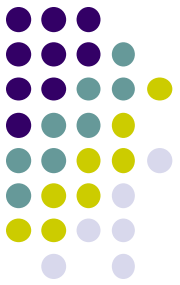
What?

- We will understand the conceptual idea of dynamic models and agent-based modelling
- We will learn how to make dynamic computer models
 - using Excel
 - using Python-based programming environment
- We will learn how to use such models to make intelligent systems that can interact knowledgeable with humans



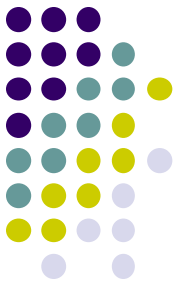
Which form?

- Lectures
- Assignments
 - weekly homework assignments
 - deadline Monday 23:59 (earlier is allowed 😊)
 - groups of 2
 - a final assignment
 - extension of lecture material
 - two phase submission (concept + final)
 - groups of 2
- Weekly online quizzes
- Brief exam about theory



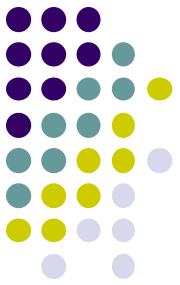
Contact hours

- Two times per week lectures
 - Wednesday 13.30-15.15
 - Friday 13.30-15.15
- Weekly support for assignments
 - 8 working groups
 - Friday 9:00 – 10:45
 - Explanation by TA's of the assignment



Explanation assignments

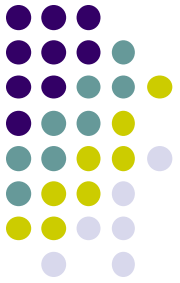
- Guidance and help
 - during support hours via TA's
- Failure of making the deadline
 - 1 point subtraction per day
- **IMPORTANT:** enroll as group (of 2) in Canvas before tomorrow noon



Course overview

Week	Theme	Lecture	Topic	Assignment
1	What is dynamic model?	1	Explanation of what Dynamic Modelling is	1. Formulating hypothesis + adapting models
		2	Modelling physiological processes	
2	Modelling social processes	3	Modelling epidemics and self-organisation	2. Performing simulations + formalization
		4	Emotion contagion + formalization	
3	Modelling behavior	5	What is behavior modelling?	3. Creating a Python-Leadsto model
		6	Delayed response and motivation-based behavior	
4	Domain models	7	-- Q&A --	
		8	Designing domain models	
5	Agent-based modelling	9	Introduction and ontologies	4. Designing and implementing a domain model
		10	Integrating domain models	
6	Analysis and support models	11	Analysis model	5. Developing an analysis model Final assignment - draft
		12	Support models + reasoning methods	
7	Wrap-up	13	Example: driving behavior	Final assignment - complete
		14	Summary	

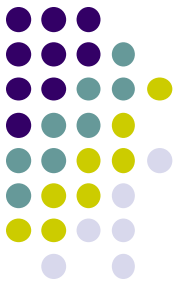
Course Material



- Reader
 - Methodological chapters:
 - Introduction to Dynamic Modelling
 - Behaviour Modelling
 - Integration of Domain Models within Agent Models
 - Methodological Aspects of Integrative Modelling
 - Integration of Logical and Numerical Modelling
 - Domain chapters:
 - Decision making, Learning
 - Honey bees, Epidemics, Emotion Contagion
 - Driving Behaviour
 - Emotion Contagion
- Slides
- Assignments

everything will be made available via Canvas (see Modules).

Grading



- The final grade will be determined by
 - results weekly assignments (A)
 - except programming-tutorial (A3), which needs to be sufficient and signed off during computer practicum
 - results final assignment (F)
 - result of exam (E)

$$\text{final grade} = 0.4 * \text{avg}(A) + 0.2 * F + 0.4 * E$$

iff all components are > 5.5 and quizzes have been done

- you are only allowed to do a resit for an assignment if you attended 5 out of 7 working groups

In case of questions / exceptions



- Related to assignments / presence
 - discuss it with one of the TA's of your working group
 - if that doesn't solve it, contact the coordinating TA Sofronia
- Related to content of course
 - ask during lecture
 - if not possible contact Nimat Ullah via Canvas

Do not send emails to multiple teachers!